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MariaDB Xpand (Cultrix) vs Traditional and NoSQL Databases: A Comparative Analysis of Distributed SQL Performance

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Abstract — *Traditional database systems are finding it difficult to provide high performance and scalability due to the explosive increase of data-intensive applications. NoSQL databases provide scalability, but consistency is frequently sacrificed. The advantages of both strategies have been combined in NewSQL systems. The architecture and performance of MariaDB Xpand, a NewSQL solution, are examined in this study. Scalability, consistency, and performance are compared with conventional MySQL and NoSQL databases. The results demonstrate that MariaDB Xpand maintains ACID compliance while achieving horizontal scalability. Nevertheless, it increases the complexity of the system. All things considered, NewSQL systems offer a practical option for contemporary large-scale applications.*

Keywords — *MariaDB Xpand, NewSQL, distributed databases, ACID transactions, horizontal scalability*

I. INTRODUCTION

Database systems that can effectively manage massive workloads are necessary due to the fast growth of data produced by contemporary applications. Due to their emphasis on vertical scaling, traditional relational databases like

MySQL have limited scalability despite offering good consistency [1].

NoSQL databases are less appropriate for transactional systems because they frequently sacrifice consistency in order to solve scalability difficulties through horizontal scaling [4]. NewSQL systems have been created to combine scalability with ACID features in order to get around these restrictions [2].

This study examines MariaDB Xpand's performance and system architecture in comparison to conventional and NoSQL databases.

II. MAIN CONTENT

A. Overview and Architecture

MariaDB Xpand, originally developed as Clustrix (also referred to as Cultrix in some tools), is designed to support distributed SQL processing in scalable environments. It is a distributed SQL database designed for scalable transactional processing. It uses a shared-nothing **architecture, where data is distributed across multiple nodes. This enables horizontal scaling** and improved performance compared to traditional MySQL systems [1].

The system supports automatic data partitioning, replication, and parallel query execution. These features ensure high availability and efficient data processing in distributed environments [2].

B. Performance and Features

MariaDB Xpand is a distributed SQL (NewSQL) database system with strong ACID compliance and support for horizontal scaling. It is appropriate for large-scale and cloud contexts since it permits SQL-based programming while distributing data across several nodes. Studies on distributed SQL systems show that this architecture outperforms conventional single-node databases in terms of availability and performance [5].

However, because of node-to-node communication and data coordinating procedures, distributed systems impose extra overhead. In comparison to traditional databases, this raises system complexity and resource consumption. Research on NewSQL systems reveals that although they increase scalability, maintaining consistency across remote contexts requires more processing power [5].

C. Comparative Analysis

Criteria	mySQL	noSQL	MariaDB Xpand
Scalability	Vertical	Horizontal	Horizontal
Consistency	Strong	Eventual	Strong
Performance	Limited	High	High
Schema	Fixed	Flexible	Relational
Complexity	Low	Medium	High

Figure 1: Comparison of Database Technologies (MySQL, NoSQL, MariaDB Xpand)

Vertical scaling limits limit the performance of standard MySQL systems in large-scale contexts, where advances in performance rely more on hardware upgrades than on distribution strategies [6]. NoSQL systems, on the other hand, use horizontal scaling to increase scalability and performance, however this frequently results in a reduction of rigorous data consistency through eventual consistency models [6].

By providing distributed scalability and robust ACID compliance, MariaDB Xpand integrates both strategies. Because of this, it is appropriate for cloud-based and enterprise applications that need both performance and dependability. Nevertheless, in comparison to MySQL and NoSQL systems, this equilibrium raises system complexity and resource consumption [5].

D. Real-World Application

Traditional web applications that need organized data and robust consistency, such as content management systems, e-commerce platforms, and enterprise systems, frequently employ MySQL.

NoSQL databases are frequently employed in large-scale applications where scalability and flexibility are more crucial than rigorous consistency, such as social media platforms, IoT systems, and real-time analytics [6].

In cloud-native and enterprise settings where scalability and relational integrity are essential, MariaDB Xpand is utilized. It is appropriate for high-availability distributed applications, SaaS platforms, and financial systems because it is based on distributed SQL concepts [5]

III. CONCLUSION

In this study, MariaDB Xpand was assessed as a NewSQL database and contrasted with conventional MySQL and NoSQL systems. The outcomes demonstrate how MariaDB Xpand effectively blends robust consistency and scalability.

It adds more complexity and resource requirements despite its benefits. The goal of future innovations should be to increase productivity without sacrificing performance. The integration of tools such as Cultrix further enhances the capability of MySQL-based systems in handling distributed workloads.

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